

This project roadmap follows an Agile methodology, organized into 12 weeks (3 months) of development, leading to a MVP (Minimum Viable Product) launch.

1. Estimated Duration

- **Total Duration:** 12 Weeks (3 Sprints of 4 weeks each).

2. Weekly Timeline

Phase	Weeks	Focus Area
	1-4	Architecture, Auth, Onboarding & AI Integration.
	5-8	Workout Engine, Tracking, & Dashboard UI.
	9-12	Notifications, Testing, Security, & Deployment.

Sprint Breakdown:

- **Weeks 1-2:** Set up cloud infrastructure (AWS/GCP), implement secure Authentication (OAuth/JWT), and build the Onboarding flow.
- **Weeks 3-4:** Integrate selected AI LLM API. Develop prompt engineering templates for fitness/nutrition plans.
- **Weeks 5-6:** Build Workout Logging system and database schema for user progress metrics.
- **Weeks 7-8:** Develop Data Visualization (Charts/Graphs) and Responsive Dashboard UI.
- **Weeks 9-10:** Develop Notification Engine (AWS SES/SendGrid integration) and trigger logic.
- **Weeks 11-12:** Security audit, GDPR compliance review, QA/Bug fixing, and Final UAT (User Acceptance Testing).

3. Milestones

- M1: Foundation Ready (Week 2):** Auth flow and user profile database schema operational.
- M2: AI Core Functional (Week 4):** First successful AI-generated personalized fitness plan delivered to frontend.
- M3: MVP Feature Complete (Week 10):** Tracking, Dashboard, and Notifications fully integrated.

4. **M4: Launch Ready (Week 12):** Security pass completed, 99.9% uptime validation, and production deployment.

4. Deliverables

- Technical:** API Documentation (Swagger/OpenAPI), Database Schema, CI/CD Pipeline.
- Product:** Secure Web Platform (MVP) featuring:
 - Onboarding survey.
 - Dashboard with personalized daily AI guidance.
 - Progress tracker (Visualization of metrics).
 - Weekly email automated summary.
- Compliance:** GDPR/CCPA Privacy Policy and "Disclaimer of Medical Advice" UI overlay.

5. Risks & Mitigation

Risk	Impact	Mitigation Strategy
AI Hallucinations	High	Implement "System Prompts" that restrict AI to safe, standardized movements and include mandatory medical disclaimers in all responses.
Data Leakage	Critical	Use AES-256 encryption at rest, TLS 1.3 in transit, and perform a third-party security penetration test before launch.
API Dependency	Medium	Implement an "Offline/Fallback" cache for routines; monitor API health with automated circuit breakers to alert users of downtime.
Scope Creep	Low	Strict adherence to the "Out of Scope" list; push any feature requests (e.g., social features) to the Version 2.0 backlog.

6. Agile Project Management Notes

- Ceremonies:** Weekly Sprint Planning, Daily Stand-ups (15 min), End-of-Sprint Retrospective.

- **Velocity Tracking:** We will use Jira or Linear to track points per sprint to adjust delivery expectations by Week 6.
- **Quality Assurance:** Automated unit testing on the AI prompt generator and manual QA for the responsive UI on desktop/mobile browsers.